How TiRex-2 makes a forecast

NX-AI · 83M parameters · xLSTM along time, attention across series · zero-shot

1. One row per series

Targets, past-only and known-ahead covariates. A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

Past only NAO index

Five metal prices go in together as five target rows.

2. Scale, squash, then patch

Each row is scaled by its own mean μ and standard deviation σ, then squashed by arcsinh; 0/1 rows are left as they are.

$$\tilde{x} = \operatorname{arcsinh}\left(\frac{x-\mu}{\sigma}\right)$$

value seen?

one patch = 32 steps

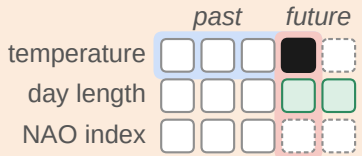
residual MLP

one token: 512 numbers

A target with a strong straight-line trend is forecast as differences.

3. Twelve blocks: time, then rows

Each block mixes along time, then across rows. Every part has a norm in front and a shortcut around it.



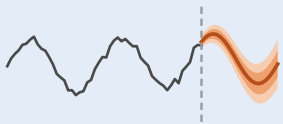
Time mixer: xLSTM along the row (mLSTM and sLSTM in turn); known-ahead rows also run backwards

Row mixer: attention down the column; targets read covariates, never the reverse

Feed forward after each mixer, token by token

4. 320 steps, one pass

Ten future patches go in as missing values. After a final norm, a head turns each into **32 steps × 9 quantiles**.



median

60% band

80% band

A fixed calibration rescales the bands, the scaling is undone, and the median is the point forecast.

Recurrence along time, attention across rows: covariates steer the targets, never the reverse.

12 blocks · width 512 · 4 heads · patches of 32 · context 2,048 steps · up to 320 steps per pass

In the notebooks: last 2,048 days or full monthly history · all covariates and the five metals together · Apache-2.0